

LARGE LANGUAGE MODELS ASSUME PEOPLE ARE MORE RATIONAL THAN WE REALLY ARE

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ABSTRACT

In order for AI systems to communicate effectively with people, they must understand how we make decisions. However, people’s decisions are not always rational, so the implicit internal models of human decision-making in Large Language Models (LLMs) must account for this. Previous empirical evidence seems to suggest that these implicit models are accurate — LLMs offer believable proxies of human behavior, acting how we expect humans would in everyday interactions. However, by comparing LLM behavior and predictions to a large dataset of human decisions, we find that this is actually not the case: when both simulating and predicting people’s choices, a suite of cutting-edge LLMs (GPT-4o & 4-Turbo, Llama-3-8B & 70B, Claude 3 Opus) assume that people are more rational than we really are. Specifically, these models deviate from human behavior and align more closely with a classic model of rational choice — expected value theory. Interestingly, people also tend to assume that other people are rational when interpreting their behavior. As a consequence, when we compare the inferences that LLMs and people draw from the decisions of others using another psychological dataset, we find that these inferences are highly correlated. Thus, the implicit decision-making models of LLMs appear to be aligned with the human expectation that other people will act rationally, rather than with how people actually act.

1 INTRODUCTION

Every day, our actions are based on decisions that reflect our internal goals and beliefs about the world. Through countless interactions with others, we are able to effortlessly predict how other people would act from their goals and beliefs, and infer others’ goals and beliefs when observing their actions. These abilities — termed forward- and inverse-modeling in cognitive science (Ho and Griffiths, 2021) — are characteristic of the implicit mental decision-making models that we form of others, and are crucial to interpersonal communication and learning (Baker et al., 2009; Lucas et al., 2014; Jara-Ettinger et al., 2020). However, while these abilities are inherent in people, the consistency and accuracy of decision-making models in Large Language Models (LLMs) is unknown. As LLMs become widely used as the basis for building AI agents that interact with or even simulate people, it is important to ask what decision-making models LLMs implicitly use: the ability to predict and interpret people’s behavior is a precursor to identifying effective ways to provide assistance, simulating the helpfulness or harmlessness of a response, and learning individuals’ values and preferences, all of which are principal to the development of safe and beneficial AI systems.

Though LLMs have become increasingly capable of conducting reasoning and conversing with humans, there is no guarantee that their implicit representations of humans align with how we behave. Methods such as Proximal Policy Optimization (Schulman et al., 2017) and Direct Preference Optimization (Rafailov et al., 2023) can be used to tune models on explicitly declared human

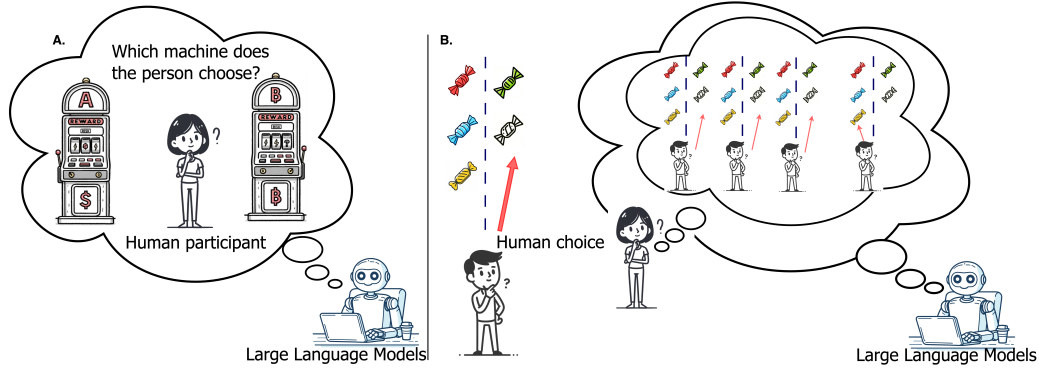


Figure 1: Two tasks we use to assess the implicit assumptions that LLMs make about human decision-making. (A) Predicting choices between gambles. Each gamble is described by the probabilities and values of different outcomes, and the goal is to predict what people will choose. (B) Inferring preferences from choices. Here, a person chooses one of many sets of objects and the goal is to infer their preferences based on that choice.

preferences (Ouyang et al., 2022), but training data such as blog posts, news articles, and books go through rounds of editing that remove logical fallacies and mistakes, leading to more “perfect” content being used for training (Cui et al., 2023; Zhou et al., 2024a). While this improves the quality of generation, it may also lead LLMs to develop mistaken impressions of how humans actually behave. In addition, many of our methods comparing LLMs to human behavior rely on human perception to measure similarity (Park et al., 2023; Jones and Bergen, 2023; Jakesch et al., 2023; Hämäläinen et al., 2023), but the psychology literature has shown that people’s perceptions of other people are flawed — we expect others to be more rational than they are (e.g., Jern et al., 2017). Thus, AI systems that act rationally can appear human-like to the naked eye, while not actually capturing human behavior.

This phenomenon is not without precedent — early economists embraced the assumption of human rationality (Smith, 1776; Mill, 1836) and built sophisticated models and policies around it (Downs, 1957; Coleman, 1994; Schelling, 1980; Dunleavy, 2014, *inter alia*), before psychologists showed just how systematic and widespread its failures were in accounting for human behavior (Tversky and Kahneman, 1974; Kahneman and Tversky, 1979). As LLM-powered systems become more widely-used, misaligned representations of humans — which can lead to mismatched beliefs and failure to follow instructions (Milli and Dragan, 2020) — pose a toll on downstream applications. But how can we study LLMs’ implicit decision-making models without being affected by the human bias to assume others act rationally?

To explore the implicit decision-making models of LLMs, we leverage two experimental paradigms from psychology — a risky choice task where people choose between gambles (Kahneman and Tversky, 1979, Figure 1A), and an inference task where people infer others’ subjective utilities after observing their decisions (Jern et al., 2017, Figure 1B). In the former, we compare choices that participants made with LLM simulations and predictions across a large dataset of over 13,000 risky decisions (Bourgin et al., 2019), while in the latter we compare the inferences drawn by LLMs and by humans over a set of 47 observations (Jern et al., 2017). These two paradigms are connected by foundational theoretical models of human decision-making, under which people develop mental models of others’ goals, utilities, and decisions, and use them to 1) construct predictions about how others will behave given their beliefs, and 2) infer what people believe based on their decisions (Baker et al., 2009; Lucas et al., 2014; Jara-Ettinger et al., 2020; Ho and Griffiths, 2021).

Through our experiments, we find that LLMs model people as highly rational decision makers. In the forward modeling task, when prompting with chain-of-thought (CoT; Wei et al., 2022), LLMs consistently predict that people act more rationally than we do. For example, GPT-4o produced a Spearman correlation of 0.94 with the rational model of choosing the maximum expected value, but humans only have a correlation of 0.48. Asking LLMs to simulate the decision with CoT also yields highly rational outcomes (0.90 for GPT-4o) that are only moderately correlated with human behavior.

In both cases, zero-shot prompting generates noisy outcomes only somewhat correlated with either rational models or humans.

In the inverse modeling task, inferences that LLMs made from peoples’ choices were also consistent with assuming that humans are rational. Across two contexts, LLMs’ inferences positively correlated with predictions from rational models, where correlations increased with model capability and reasoning (0.20 for Llama-3-8B zero-shot; 0.95 for GPT-4o CoT). Interestingly, psychology literature finds that people, despite deviating from rationality in their own choices, make inferences from the behavior of others as if they were rational (Baker et al., 2009; Lucas et al., 2014; Jern et al., 2017; Jara-Ettinger et al., 2020). Thus, the inferences drawn from others’ decisions by people and LLMs should actually be similar. We find support for this: Inferences made by LLMs are highly correlated with the same inferences made by people ($\rho = 0.97$ with GPT-4o CoT). Thus, while LLMs are not accurate at simulating or predicting human behavior, LLMs’ incorrect assumption that people are more rational aligns with the assumption that people also make when interpreting another’s behavior.

The remainder of the paper is organized as follows. In Section 2, we introduce related work across LLMs, alignment, and decision-making in cognitive science. Section 3 and Section 4 describe the forward and inverse modeling experiments and results, detailing our analyses and the relationships between LLMs, humans, and existing rational theories. Finally, in Section 5 we discuss insights including new challenges for simulating humans using LLMs, misspecification of “alignment” in decision-making, and potential implications for training LLMs using different paradigms.

2 RELATED WORK

2.1 ALIGNING LARGE LANGUAGE MODELS WITH HUMANS

Large Language Models are typically aligned with human preferences through Reinforcement Learning from Human Feedback (RLHF) (Bai et al., 2022; Ouyang et al., 2022). Training with human preference data has been shown to enhance reasoning (Havrilla et al., 2024), and the impressive capacities of resulting models (e.g., Bubeck et al., 2023) have sparked interest across various fields in using them to model (e.g., Binz and Schulz, 2023a; Macmillan-Scott and Musolesi, 2024) and simulate (e.g., Park et al., 2022; Argyle et al., 2023; Liu et al., 2023; Salewski et al., 2024) human behavior. However, these models still exhibit biases and hallucinations (e.g., Jiang et al., 2024; Bai et al., 2024; Anwar et al., 2024), and may not be adept at capturing trade-offs in human behavior (Liu et al., 2024b; Coletta et al., 2024) or situations with information asymmetry (Zhou et al., 2024b).

Methods from cognitive science are increasingly being used to study LLMs (Coda-Forno et al., 2024; Liu et al., 2024b; Binz and Schulz, 2023b; Liu et al., 2024a). A particularly contentious debate is whether LLMs exhibit Theory of Mind, the ability to model others’ mental states which may be different than their own (Premack and Woodruff, 1978). Several studies have shown evidence for (Kosinski, 2023) and against (Sap et al., 2022; Ullman, 2023) Theory of Mind, including studies that develop more rigorous evaluation methods via procedural generation (Gandhi et al., 2024) and adversarial examples (Shapira et al., 2024). Our analysis provides a quantitative approach to engaging with this debate, as inverse decision-making can be considered a specific form of Theory of Mind.

2.2 FORWARD AND INVERSE MODELS OF HUMAN DECISION-MAKING

One of the most classic and extensively studied problems in decision-making is the risky choice task (Kahneman and Tversky, 1979; Edwards, 1954; Bourgin et al., 2019), where people choose among gambles with different outcome probabilities and payoffs. The rational action is to choose the gamble with the highest expected value, calculated by summing the product of the probabilities and the values of the outcomes. On this task, humans have been described as deviating from rationality in a fourfold pattern: risk seeking for small probability gains, risk averse for small probability losses, risk averse for moderate or large probability gains, and risk seeking for moderate or large probability losses (Kahneman and Tversky, 1979). Other studies have shown that humans tend to act in accordance with bounded rationality, where rationality is traded-off with mental effort, information, and time (Evans et al., 2015; Alanqary et al., 2021).

Peterson et al. (2021) collected the `choices13k` dataset, a large human dataset with over 13,000 risky choice problems, and showed that people’s decisions in this setting could be captured by simple